



Brief paper

Information-theoretic multi-time-scale partially observable systems with inspiration from leukemia treatment[☆]Margaret P. Chapman^{a,*}, Emily Jensen^{b,1}, Steven M. Chan^c, Laurent Lessard^d^a Edward S. Rogers Sr. Department of Electrical and Computer Engineering, University of Toronto, 10 King's College Road, Toronto, ON, M5S 3G8, Canada^b Department of Electrical, Computer and Energy Engineering, University of Colorado Boulder, 425 UCB, Boulder, CO, 80309, United States^c Department of Medical Biophysics, Faculty of Medicine, University of Toronto, Princess Margaret Cancer Research Tower, MaRS Centre, 101 College Street, Toronto, ON, M5G 1L7, Canada^d Department of Mechanical and Industrial Engineering, Northeastern University, 334 Snell Engineering Center, 360 Huntington Avenue, Boston, MA, 02115, United States

ARTICLE INFO

Article history:

Received 26 April 2022

Received in revised form 12 July 2023

Accepted 6 January 2024

Available online 5 February 2024

Keywords:

Stochastic control

Nonlinear systems

Partially observable systems

System identification

Biomedical systems

ABSTRACT

We study a partially observable nonlinear stochastic system with unknown parameters, where the given time scales of the states and measurements may be distinct. The proposed setting is inspired by disease management, particularly leukemia.

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1. Introduction

A patient with a disease may be viewed as a stochastic system that evolves in response to control inputs, e.g., drug doses. Aspects of the patient can be measured, e.g., a tumor can be scanned or a blood sample can be taken. Controls and measurements typically operate on different time scales; for example, measurements may be weekly, and there may be days when no drug is taken. While there is knowledge about the underlying biochemical processes, such processes may be nonlinear, noisy, and may vary between patients. In many medical settings (e.g., leukemia treatment, diabetes treatment, and blood pressure management), the doses of drugs require adjustments for the purpose of regulation (e.g., keeping white blood cells, glucose, and blood pressure, respectively, within specific ranges). From a mathematical point of view, it is unclear how to unify the above

properties (partial observability, nonlinear stochastic dynamics with uncertain parameters, and multiple time scales) into a rigorous control-theoretic framework. Our aim is to develop such a unifying framework.

We consider a discrete-time partially observable stochastic system that differs from a standard set-up. The system has an unknown deterministic parameter vector θ , representing interpatient variability. Furthermore, the states and measurements may operate on different time scales, the dynamics and measurement equations may be nonlinear, the process and measurement noise may have unbounded support, and the spaces of states, controls, and measurements do not have finite cardinality. Working in the setting above, we aim to develop a pathway that provides a control policy with awareness about the uncertainty in an estimate for θ and sensitivity of the state relative to θ . A theoretical foundation must be developed before approximations can be investigated to permit computation.

Our contribution is of a conceptual nature. We offer a theoretical framework that unifies the following key characteristics: partial observability, uncertain parameters, multiple time scales, and nonlinear stochastic dynamics (Fig. 1). First, we devise a theoretical approach to handle the unknown parameters, building on concepts from optimum experiment design and dual control (Section 3). Second, we represent the system as a multi-time-scale partially observable (PO) Markov decision process (MDP)

[☆] The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Maria Prandini under the direction of Editor Sophie Tarbouriech.

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¹ This work was completed while E. Jensen was a postdoctoral researcher at Northeastern University.

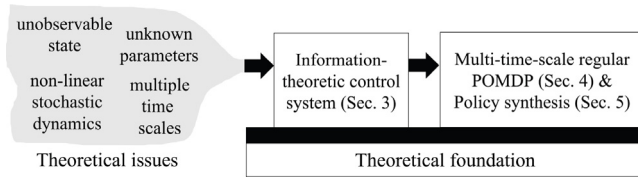


Fig. 1. A high-level illustration of this work.

with information-theoretic cost functions and show that it enjoys regularity properties under some assumptions (Section 4). Third, we provide conditions that guarantee the existence of an optimal policy for a belief-space MDP corresponding to the POMDP (Section 5). Our main contribution is to reduce the complications that arise from the unobservable state, unknown parameters, multiple time scales, and nonlinear stochastic dynamics to a form that admits a mathematical solution. We exemplify the assumptions in the context of a leukemia treatment model in Section 6. Although our presented framework does not solve the curse of dimensionality, it does provide a launching point for bridging the gap between theory and practice, as further described in our conclusion (Section 7).

1.1. Related literature

The theory is related to three bodies of literature: POMDPs, optimum experiment design, and stochastic adaptive and dual control. POMDPs with Borel state and control spaces have been studied since at least the 1970s (Rhenius, 1974; Yushkevich, 1976) (Bertsekas & Shreve, 1996, Ch. 10) (Hernández-Lerma, 1989, Ch. 4). The state-of-the-art conditions that facilitate policy synthesis for these general POMDPs were developed recently (Feinberg, Kasyanov, & Zgurovsky, 2016, 2022; Kara, Saldi, & Yüksel, 2019). We propose and study a POMDP problem with unknown parameters and information-theoretic costs, where the time scales of the measurements and states may be distinct. The different time scales in the setting of unknown parameters pose difficulties due to the irregular updates to the parameter estimates and posterior distributions. We offer a reformulation that uses measure-theoretic first principles and regularity conditions proposed by Feinberg et al. (2016) and Kara et al. (2019). The author of Hernández-Lerma (1989, Ch. 4.5) studies POMDPs with unknown parameters, continuous bounded cost functions, and weakly continuous bounded dynamics for posterior distributions. However, we consider cost functions that are lower semicontinuous and bounded below. We do not assume weakly continuous bounded dynamics for posterior distributions in view of a counterexample (Feinberg et al., 2016, Ex. 4.2). We propose stage costs that penalize poor quality of information from measurements using the Fisher information matrix from optimum experiment design (to be discussed). To accommodate these information-theoretic costs, we assume Euclidean spaces of states, parameters, measurements, and measurement noise, and we make assumptions about the existence and continuity of Jacobians.

An optimum experiment design (OED) problem is a type of optimal control problem (Sager, 2013). An example OED problem is to optimize the planned measurement times by maximizing the quality of information, which can be quantified using the Fisher information matrix (FIM) (Erdal, Plaxco, Gerson, Kippin, & Hespanha, 2021; Nimmegeers, Bhonsale, Telen, & van Impe, 2020; Pronzato, 2008; Sager, 2013). Then, an optimal control problem can be solved using the parameter estimate (Sager, 2013, Eq. (3.3)) (Jost, 2020, Algorithm 2, pp. 79–81). In clinical practice, the planned measurement times are determined by the

patient's schedule, which may be infeasible to adjust. Thus, we are not concerned with optimizing the planned measurement times. The OED literature includes measurement noise but often neglects process noise (Erdal et al., 2021; Jost, 2020; La, Potschka, Schlöder, & Bock, 2017; Sager, 2013); an exception is Telen et al. (2013). We include both measurement and process noise; the latter plays numerous roles in biochemical systems (Eling, Morgan, & Marioni, 2019; Rao, Wolf, & Arkin, 2002). The techniques in Telen et al. (2013) rely on linear approximations and a Riccati differential equation to approximate the covariance of the estimated state at the final time. In contrast, we leverage the theory of discrete-time nonlinear POMDPs for policy synthesis. Our objective evaluates the stage-wise performance of the states and controls and an FIM-based criterion.

This paper has connections to stochastic adaptive and dual control, which are related to OED. Separating the problems of parameter estimation and controller design characterizes classical adaptive control schemes, e.g., self-tuning regulators (Åström & Wittenmark, 1995, p. 22). In general, separating the two problems implies that parameter uncertainties are neglected in controller synthesis, and hence, dual control methods have been developed to tackle the two problems simultaneously (Åström & Wittenmark, 1995, pp. 22–24). A recent review of stochastic dual control is provided by Mesbah (2018), and a concise summary is provided by Hewing, Wabersich, Menner, and Zeilinger (2020, p. 276). We discuss some recent papers on dual control, which involve systems with unknown parameters (exception: Feng & Houska, 2018). Model predictive controllers for linear systems with process noise (Bavdekar, Ehlinger, Gidon, & Mesbah, 2016) and nonlinear systems without process noise (La et al., 2017) have been developed, where the objective is a sum of a performance metric and an FIM-based metric. The authors of La et al. (2017) also assume initial state distributions with bounded support. Heirung et al. develop a model predictive controller using a quadratically-constrained quadratic program to minimize the predicted mean-squared output error; the system is a linear regression model with normally distributed noise and parameters (Heirung, Ydstie, & Foss, 2017). Model predictive controllers with information matrix-based constraints have been proposed for linear systems with process noise (Larsson, Ebadat, Rojas, Bombois, & Hjalmarsson, 2016) and nonlinear systems without process noise (Telen, Houska, Vallerio, Logist, & van Impe, 2017). The controller in Larsson et al. (2016) incorporates different time scales for the measurements and controls. Feng and Houska develop a real-time model predictive control algorithm for a nonlinear system to optimize a sum of a performance metric and an approximation for the average loss of optimality due to poor future state estimates (Feng & Houska, 2018). They assume that the measurement and process noise have bounded support to facilitate the latter approximation, which employs an extended Kalman filter (Feng & Houska, 2018). In contrast to the above papers, we provide a theoretical study of a partially observable multi-time-scale nonlinear system with unknown parameters, where the measurement and process noise may have unbounded support. We include an FIM-based criterion in the objective rather than as a constraint to avoid potential feasibility issues.

1.2. Notation

$\mathbb{R}$ is the real line. $\mathbb{R}^* := \mathbb{R} \cup \{-\infty, +\infty\}$ is the extended real line. $\mathbb{R}^n$ is n -dimensional Euclidean space, $\mathbb{R}_+^n := \{x \in \mathbb{R}^n : x_i > 0, i = 1, 2, \dots, n\}$, and $\mathbb{R}_+ := \mathbb{R}_+^1$. $\mathbb{N}$ is the set of natural numbers. If $f : \mathbb{R}^n \times \mathbb{R}^l \rightarrow \mathbb{R}^m$ with $(x, y) \mapsto f(x, y)$, then $\frac{\partial f(a,b)}{\partial x} \in \mathbb{R}^{m \times n}$ is the Jacobian matrix of f with respect to x evaluated at $(a, b) \in \mathbb{R}^n \times \mathbb{R}^l$. The Euclidean norm is $\|\cdot\|$. If $A \in \mathbb{R}^{m \times n}$, we define $(*)^T A := A^T A$. We define $\varphi_\epsilon : \mathbb{R}^l \rightarrow \mathbb{R}_+^l$ by $\varphi_\epsilon(x) :=$

$(\max\{x_1, \epsilon\}, \max\{x_2, \epsilon\}, \dots, \max\{x_l, \epsilon\})^\top$, where $\epsilon \in \mathbb{R}_+$ is near zero. If M is a separable metrizable space, then $\mathcal{B}_M$ is the Borel sigma algebra on M , $\mathcal{P}_M$ is the collection of probability measures on $(M, \mathcal{B}_M)$ with the weak topology, and $\|\cdot\|_{TV}$ is the total variation norm on $\mathcal{P}_M$. If $p_i \in \mathcal{P}_M$ for $i = 1, 2$, the total variation norm of $p_1 - p_2$ is $\|p_1 - p_2\|_{TV} = 2 \sup_{M \in \mathcal{B}_M} |p_1(M) - p_2(M)|$. We use the notation $\underline{M}$ to denote a Borel set, i.e., $\underline{M} \in \mathcal{B}_M$, following the notation from Bertsekas and Shreve (1996). If $y \in M$, then δ_y is the Dirac measure in $\mathcal{P}_M$ concentrated at y . If A and B are two sets, then $A \setminus B$ means A set minus B . Note the following abbreviations: l.s.c. = lower semicontinuous, b.b. = bounded below, measurable = Borel-measurable, and Rmk. = Remark.

1.3. Preliminaries

To keep the work self-contained, we recall some principles mostly from Bertsekas and Shreve (1996). Related material is available from Ash (1972), Hernández-Lerma (1989), Hernández-Lerma and Lasserre (1996) and Hinderer (1970). X and Y are separable metrizable spaces.

Remark 1 (Weak Convergence). Let $(p_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{P}_X$ and $p \in \mathcal{P}_X$. The (weak) convergence of $p_n \rightarrow p$ in $\mathcal{P}_X$ is equivalent to $\int_X f dp_n \rightarrow \int_X f dp$ in $\mathbb{R}$ for every continuous bounded function $f : X \rightarrow \mathbb{R}$ (Bertsekas & Shreve, 1996, Prop. 7.21).

Remark 2 (Stochastic Kernels). A measurable stochastic kernel $q(dy|x)$ on Y given X is a family of elements of $\mathcal{P}_Y$ parametrized by elements of X , where the map $v : X \rightarrow \mathcal{P}_Y$ defined by $v(x) := q(\cdot|x)$ is measurable (Bertsekas & Shreve, 1996, Def. 7.12). If v is weakly continuous, i.e., $x_n \rightarrow x$ in X implies $q(\cdot|x_n) \rightarrow q(\cdot|x)$ in $\mathcal{P}_Y$, then q is called *weakly continuous*. If v is continuous in total variation, i.e., $x_n \rightarrow x$ in X implies $\|q(\cdot|x_n) - q(\cdot|x)\|_{TV} \rightarrow 0$, then q is called *continuous in total variation*.

Remark 3 (Some Continuity Facts). Let q be a weakly continuous stochastic kernel on Y given X , $f : X \times Y \rightarrow \mathbb{R}^*$ be measurable and b.b., and $g : X \rightarrow \mathbb{R}^*$ be defined by $g(x) := \int_Y f(x, y) q(dy|x)$. If f is continuous and bounded, then g is continuous and bounded (Bertsekas & Shreve, 1996, Prop. 7.30). If f is l.s.c. (and b.b.), then g is l.s.c. and b.b. (Bertsekas & Shreve, 1996, Prop. 7.31). The map $\ell : Y \rightarrow \mathcal{P}_Y$ defined by $\ell(y) := \delta_y$ is weakly continuous (Bertsekas & Shreve, 1996, Cor. 7.21.1). The map $\sigma : \mathcal{P}_X \times \mathcal{P}_Y \rightarrow \mathcal{P}_{X \times Y}$ defined by $\sigma(p_1, p_2) := p_1 p_2$ is weakly continuous (Bertsekas & Shreve, 1996, Lemma 7.12). The construction of the product $p_1 p_2 \in \mathcal{P}_{X \times Y}$ of $p_1 \in \mathcal{P}_X$ and $p_2 \in \mathcal{P}_Y$ is provided by Ash (1972, Cor. 2.6.3).

Remark 4 (Kernel δ_f). If $f : X \rightarrow Y$ is measurable, we define δ_f by $\delta_f(x) := \ell(f(x))$ for all $x \in X$, where ℓ is the Dirac measure map defined in Remark 3. δ_f is a measurable stochastic kernel on Y given X due to the measurability of f and the (weak) continuity of ℓ .

Remark 5 (Measurable Selection). Assume that Y is compact. Let $f : X \times Y \rightarrow \mathbb{R}^*$ be l.s.c. and b.b., and define $f^* : X \rightarrow \mathbb{R}^*$ by $f^*(x) := \inf_{y \in Y} f(x, y)$. Then, f^* is l.s.c. and b.b., and there exists a measurable function $\kappa : X \rightarrow Y$ such that $f(x, \kappa(x)) = f^*(x)$ for all $x \in X$ (Bertsekas & Shreve, 1996, Prop. 7.33).

2. System description

We consider a discrete-time stochastic system that differs from a standard one. There is an unknown deterministic parameter vector, there may be different time scales, and the state is not observable. The system takes the following form:

$$\begin{aligned} x_{t+1} &= f_t(x_t, u_t, d_t; \theta), & t \in \mathbb{T}, \\ y_t &= h_t(x_t; \theta) + w_t, & t \in \mathbb{T}_y, \end{aligned} \quad (1)$$

where $x_t \in \mathbb{S}$ is a state, $y_t \in \mathbb{Y} := \mathbb{R}^m$ is a measurement, $u_t \in \mathbb{C}$ is a control, $d_t \in \mathbb{D}$ is a process noise realization, $w_t \in \mathbb{Y}$ is a measurement noise realization, and $\theta \in \mathbb{P} := \mathbb{R}_+^p$ is a parameter vector whose true value is unknown. An initial estimate for θ is available. The state space $\mathbb{S} \in \mathcal{B}_{\mathbb{R}^n}$ is a nonempty Euclidean space. The control space $\mathbb{C}$ and the process noise space $\mathbb{D}$ are nonempty Borel spaces (Bertsekas & Shreve, 1996, Def. 7.7). The states evolve on $\mathbb{T} := \{0, 1, \dots, N-1\}$, a time horizon of length $N \in \mathbb{N}$. The measurements evolve on a nonempty subset $\mathbb{T}_y$ of $\mathbb{T}_N := \mathbb{T} \cup \{N\}$. The controls may be optimized on a nonempty subset $\mathbb{T}_u$ of $\mathbb{T}$. If $t \in \mathbb{T} \setminus \mathbb{T}_u$, then u_t is assigned a given default value $\mathbf{u} \in \mathbb{C}$. The functions $f_t : \mathbb{S} \times \mathbb{C} \times \mathbb{D} \times \mathbb{P} \rightarrow \mathbb{S}$ for every $t \in \mathbb{T}$ and $h_t : \mathbb{S} \times \mathbb{P} \rightarrow \mathbb{Y}$ for every $t \in \mathbb{T}_y$ are measurable. These functions and the horizons $\mathbb{T}$, $\mathbb{T}_y$, and $\mathbb{T}_u$ are given. The quantities x_t , u_t , d_t , y_t , and w_t in (1) are realizations of random objects X_t , U_t , D_t , Y_t , and W_t , respectively. The random objects X_0 , D_t for every $t \in \mathbb{T}$, and W_t for every $t \in \mathbb{T}_y$ are independent. D_t for any $t \in \mathbb{T}$ and W_t for any $t \in \mathbb{T}_y$ do not depend on θ . The (prior) distribution of X_0 is given, and the distributions of D_t and W_t are given when these random objects are defined. (Assuming knowledge of such distributions is typical in research about partially observable systems.) We use the following notations:

- If $t \in \mathbb{T} \setminus \mathbb{T}_y$, then the distribution of D_t is $\mu_t \in \mathcal{P}_{\mathbb{D}}$.
- If $t \in \mathbb{T}_y$, then the distribution of W_t is $\rho_t \in \mathcal{P}_{\mathbb{Y}}$.
- If $t \in \mathbb{T}_y \setminus \{N\}$, then the distribution of (D_t, W_t) is $\nu_t \in \mathcal{P}_{\mathbb{D} \times \mathbb{Y}}$.

ρ_t and ν_t are related by $\rho_t(\mathbb{Y}) = \nu_t(\mathbb{D} \times \mathbb{Y})$ for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$ and $t \in \mathbb{T}_y \setminus \{N\}$. The system (1) has a stage cost function $\hat{c}_t : \mathbb{S} \times \mathbb{C} \times \mathbb{P} \rightarrow \mathbb{R}^*$ for every $t \in \mathbb{T}$ and a terminal cost function $\hat{c}_N : \mathbb{S} \times \mathbb{P} \rightarrow \mathbb{R}^*$, which may depend on θ . These functions assess the performance of the states and controls. We invoke the following assumption.

Assumption 1 (About f_t , $\hat{c}_t$, $\mathbb{C}$, h_t , ρ_t). We assume:

- f_t is continuous for every $t \in \mathbb{T}$;
- $\hat{c}_t$ is l.s.c. and bounded below for every $t \in \mathbb{T}_N$;
- $\mathbb{C}$ is compact;
- h_t is continuous for every $t \in \mathbb{T}_y$;
- ρ_t admits a continuous (nonnegative) density with respect to Lebesgue measure in $\mathcal{P}_{\mathbb{Y}}$ for every $t \in \mathbb{T}_y$;
- $f_t(x, u, d; \theta)$ is differentiable in x and θ for every $t \in \mathbb{T}$; $h_t(x; \theta)$ is differentiable in x and θ for every $t \in \mathbb{T}_y$; the associated Jacobians are continuous.

We abbreviate Assumption 1 as A1. Parts (a)–(c) of A1 are standard conditions to help ensure the existence of an optimal policy when the objective is an expected cumulative cost and the state is observable (Bertsekas & Shreve, 1996, Def. 8.7). Parts (d)–(e) will help guarantee the regularity of observation kernels. Part (f) will facilitate estimating θ using measurements and assessing the quality of the measurements. In particular, our leukemia treatment model satisfies A1 (Section 6).

3. Information-theoretic control system

Toward designing a control policy for (1), here we provide a pathway for estimating θ and managing potential differences between $\mathbb{T}_N$ and $\mathbb{T}_y$. First, we specify cost functions to assess the quality of measurements using concepts from optimum experiment design. Second, we define an enlarged system to record these information-theoretic costs. We conclude the section by studying the system's properties in Theorem 1.

3.1. Formulating information-theoretic costs

We would like to extract useful information from measurements to inform the estimation of θ . The OED literature assesses the usefulness of information through the Fisher information matrix (FIM). For the system (1) and a state trajectory $(x_0, x_1, \dots, x_N) \in \mathbb{S}^{N+1}$, the FIM $\mathcal{F} \in \mathbb{R}^{p \times p}$ is defined by $\mathcal{F} := \sum_{t \in \mathbb{T}_y} \mathcal{F}_t((x_t^\top, \xi_t^\top, \theta^\top)^\top)$ (Sager, 2013, Def. 3.2), where each term

$$\mathcal{F}_t((x_t^\top, \xi_t^\top, \theta^\top)^\top) := (*)^\top \left(\frac{dh_t(x_t; \theta)}{d\theta} \right), \quad t \in \mathbb{T}_y, \quad (2a)$$

depends on the output sensitivity matrix (Keesman, 2011, Eq. (6.101))

$$\frac{dh_t(x_t; \theta)}{d\theta} := \frac{\partial h_t(x_t; \theta)}{\partial x} \cdot \xi_t + \frac{\partial h_t(x_t; \theta)}{\partial \theta}, \quad t \in \mathbb{T}_y, \quad (2b)$$

and the state sensitivity matrix $\xi_t := \frac{dx_t}{d\theta} \in \mathbb{R}^{n \times p}$. The output sensitivity matrix (2b) is the total derivative of the predicted measurement $h_t(x_t; \theta)$ with respect to θ . The ξ_t -dynamics are defined by²

$$\xi_{t+1} = \phi_t(\xi_t, x_t, u_t, d_t; \theta), \quad t \in \mathbb{T}, \quad (3a)$$

where

$$\phi_t(\xi, x, u, d; \theta) := \frac{\partial f_t(x, u, d; \theta)}{\partial x} \cdot \xi + \frac{\partial f_t(x, u, d; \theta)}{\partial \theta}. \quad (3b)$$

If θ does not include the initial state x_0 , then ξ_0 is the zero matrix in $\mathbb{R}^{n \times p}$. If θ does include the initial state, then ξ_0 consists of an identity matrix and a zero matrix.

We would like the size of the FIM $\mathcal{F}$, e.g., its trace or determinant, to be large (Sager, 2013, Def. 3.4). Statistical background underlying $\mathcal{F}$ can be found in Ljung (1999). We content ourselves with an intuitive explanation. A large $\text{trace}(\mathcal{F})$ means that the predicted measurement $h_t(x_t; \theta)$ varies a large amount with respect to θ . Therefore, different values of θ correspond to different measurements, which facilitates the estimation of the true θ from the measurements. Defining an augmented state that includes the state sensitivity matrix ξ_t is useful for assessing FIM-based criteria in optimal control problems (Larsson et al., 2016; Sager, 2013).

Eqs. (2)–(3) depend on θ , whose true value is unknown. Hence, one evaluates these equations using an estimate for θ , which we denote by $\hat{\theta}_t \in \mathbb{P}$. Consequently, we must track the evolution of the estimates, which is an aspect of dual control (Åström & Wittenmark, 1995, p. 23) (Hernández-Lerma, 1989, Ch. 2.5).

In view of the above, we consider an augmented state $\chi_t := (x_t^\top, \xi_t^\top, \hat{\theta}_t^\top)^\top$ with values in $\mathbb{X} := \mathbb{S} \times \mathbb{R}^{n \times p} \times \mathbb{P}$. We specify a cost function $\bar{c}_t : \mathbb{X} \rightarrow \mathbb{R}$, which aims to penalize a small trace of $\mathcal{F}_t(\chi)$ (2) for every $\chi = (x^\top, \xi^\top, \hat{\theta}^\top)^\top \in \mathbb{X}$,

$$\bar{c}_t(\chi) := -\min\{\text{trace}(\mathcal{F}_t(\chi)), b\}, \quad t \in \mathbb{T}_y, \quad (4)$$

where $b \in \mathbb{R}_+$ is a large constant. One selects b empirically so that $\min\{\text{trace}(\mathcal{F}_t(\chi)), b\} = \text{trace}(\mathcal{F}_t(\chi))$ for all $\chi \in \mathbb{X}$ of practical interest. b is needed so that $\bar{c}_t$ is b.b.

² The definition (3) uses the mild assumption that the process noise does not depend on θ . Moreover, the definition (3) neglects the dependency of u_t on θ . This simplification is standard, e.g., see Sager (2013, Def. 3.1) and Keesman (2011, Eq. (6.100)) for continuous-time examples, because how u_t depends on the information available up to time t , i.e., $i_t := (y_0, u_0, \dots, y_{t-1}, u_{t-1}, y_t)$, is not known *a priori*. An alternative is to restrict oneself to controls of the form $u_t = K_t(i_t)$, where K_t is a linear combination of differentiable functions, e.g., polynomials, so that $\frac{du_t}{d\theta}$ can be evaluated and included in the ξ_t -dynamics (3). While this alternative is out of scope, it may be interesting for future study.

Using $\bar{c}_t$ (4) and $\hat{c}_t$ (Section 2), we define stage cost functions $c_t : \mathbb{X} \times \mathbb{C} \rightarrow \mathbb{R}^*$ for all $t \in \mathbb{T}$ that include both information-theoretic and performance-based criteria:

$$c_t(\chi, u) := \begin{cases} \hat{c}_t(x, u; \hat{\theta}) + \lambda \bar{c}_t(\chi), & \text{if } t \in \mathbb{T}_y \setminus \{N\}, \\ \hat{c}_t(x, u; \hat{\theta}), & \text{if } t \in \mathbb{T} \setminus \mathbb{T}_y. \end{cases} \quad (5a)$$

$\lambda \in \mathbb{R}_+$ is chosen *a priori* based on the relative importance of the two types of criteria. Similarly, we define a terminal cost function $c_N : \mathbb{X} \rightarrow \mathbb{R}^*$ by

$$c_N(\chi) := \begin{cases} \hat{c}_N(x; \hat{\theta}) + \lambda \bar{c}_N(\chi), & \text{if } N \in \mathbb{T}_y, \\ \hat{c}_N(x; \hat{\theta}), & \text{if } N \notin \mathbb{T}_y. \end{cases} \quad (5b)$$

A terminal cost is important for leukemia treatment because oncologists specify that the concentration of neutrophils should be within particular bounds by the end of a treatment cycle (to be exemplified in Section 6).

An advantage of incorporating the FIM into a cost function, as in (5), is that this choice avoids feasibility issues, which may arise if the FIM is incorporated into a constraint. However, λ requires tuning. The authors of Feng and Houska (2018) propose a trade-off term so that their objective penalizes an expected loss of optimality, if the noise has sufficiently small bounded support (Feng & Houska, 2018, Th. 1). In contrast, we permit unbounded noise. To formulate an optimal control problem using the cost functions (5), we will define the dynamics of χ_t in the next subsection, starting with the dynamics of the parameter estimate $\hat{\theta}_t$.

3.2. Defining χ_t -dynamics

We define the dynamics of $\hat{\theta}_t \in \mathbb{P}$ for every $t \in \mathbb{T}$ by

$$\hat{\theta}_{t+1} = g_t(y_t, x_t; \hat{\theta}_t) := \begin{cases} \bar{g}_t(y_t, x_t; \hat{\theta}_t), & \text{if } t \in \mathbb{T}_y \setminus \{N\}, \\ \hat{\theta}_t, & \text{if } t \in \mathbb{T} \setminus \mathbb{T}_y, \end{cases} \quad (6)$$

where $\hat{\theta}_0$ is an initial guess for θ . For every $t \in \mathbb{T}_y \setminus \{N\}$, we define $\bar{g}_t$ using a gradient-based procedure (Keesman, 2011, Eq. (5.73)) adapted to ensure that $\bar{g}_t$ is $\mathbb{P}$ -valued:

$$\bar{g}_t(y, x; \theta) := \varphi_\epsilon(\theta - \alpha_t \cdot L_t(x; \theta)^{-1} \cdot D_\theta \varepsilon_t(y, x; \theta)). \quad (7)$$

In (7), $\varphi_\epsilon : \mathbb{R}^p \rightarrow \mathbb{P}$ is continuous (Section 1.2), $\alpha_t \in \mathbb{R}_+$ is a step size, $D_\theta \varepsilon_t(y, x; \theta) \in \mathbb{R}^p$ is the gradient of the mean-squared measurement errors $\varepsilon_t(y, x; \theta) := \|y - h_t(x; \theta)\|^2$ with respect to θ , and $L_t(x; \theta) \in \mathbb{R}^{p \times p}$ is symmetric positive definite (to be specified). We may choose $L_t(x; \theta)$ to be the identity matrix I_p so that (7) resembles gradient descent. Otherwise, we use a Gauss–Newton update with a regularization term $\gamma \in \mathbb{R}_+$ (Keesman, 2011, Eq. (5.79)):

$$L_t(x; \theta) = 2(*)^\top \left(\frac{\partial h_t(x; \theta)}{\partial \theta} \right) + \gamma I_p. \quad (8)$$

The χ_t -dynamics depend on an output equation H_t and a state update equation F_t . Using h_t (1) and an arbitrary vector $v \in \mathbb{Y}$, we define $H_t : \mathbb{X} \times \mathbb{Y} \rightarrow \mathbb{Y}$ by

$$H_t(\chi, w) := \begin{cases} h_t(x; \hat{\theta}) + w, & \text{if } t \in \mathbb{T}_y, \\ v, & \text{if } t \in \mathbb{T}_N \setminus \mathbb{T}_y. \end{cases} \quad (9)$$

v is a “dummy” vector that will be useful for writing the χ_t -dynamics concisely and for analyzing these dynamics. We define $F_t : \mathbb{X} \times \mathbb{C} \times \mathbb{D} \times \mathbb{Y} \rightarrow \mathbb{X}$ by

$$F_t(\chi, u, d, w) := \bar{F}_t(\chi, u, d, H_t(\chi, w)), \quad t \in \mathbb{T}, \quad (10)$$

where $\bar{F}_t$ depends on f_t (1), ϕ_t (3), and g_t (6) as follows:

$$\bar{F}_t \left(\begin{bmatrix} x \\ \xi \\ \hat{\theta} \end{bmatrix}, u, d, y \right) := \begin{bmatrix} f_t(x, u, d; \hat{\theta}) \\ \phi_t(\xi, x, u, d; \hat{\theta}) \\ g_t(y, x; \hat{\theta}) \end{bmatrix}, \quad t \in \mathbb{T}. \quad (11)$$

Then, the χ_t -dynamics are given by

$$\begin{aligned}\chi_{t+1} &= F_t(\chi_t, u_t, d_t, w_t), \quad t \in \mathbb{T}, \\ y_t &= H_t(\chi_t, w_t), \quad t \in \mathbb{T}_N,\end{aligned}\quad (12)$$

where w_t is a realization of W_t if $t \in \mathbb{T}_y$, and $w_t = v$ if $t \in \mathbb{T}_N \setminus \mathbb{T}_y$. v does not affect χ_{t+1} for any $t \in \mathbb{T}$. For every $t \in \mathbb{T} \setminus \mathbb{T}_y$ and $(\chi, u, d, w) \in \mathbb{X} \times \mathbb{C} \times \mathbb{D} \times \mathbb{Y}$, we have that $F_t(\chi, u, d, w) = G_t(\chi, u, d)$, where

$$G_t(\chi, u, d) := \begin{bmatrix} f_t(x, u, d; \hat{\theta}) \\ \phi_t(\xi, x, u, d; \hat{\theta}) \\ \hat{\theta} \end{bmatrix}, \quad t \in \mathbb{T} \setminus \mathbb{T}_y. \quad (13)$$

$\mathbf{p} \in \mathcal{P}_{\mathbb{X}}$ denotes the prior distribution of the realizations $\chi_0 \in \mathbb{X}$. $\mathbf{p}$ depends on the distribution of X_0 and the values of ξ_0 and $\hat{\theta}_0$ (see just below (3b) and (6), respectively). Let us now study the costs and dynamics presented above.

Theorem 1 (Regularity of c_t , F_t , and G_t). *Let A1 hold. Then, c_t (5) is l.s.c. and b.b. for every $t \in \mathbb{T}_N$, and F_t (10) is continuous for every $t \in \mathbb{T}$. In particular, G_t (13) is continuous for every $t \in \mathbb{T} \setminus \mathbb{T}_y$.*

Proof. The first property holds because c_t is a sum of l.s.c. and b.b. functions, $\hat{c}_t$ or $\hat{c}_t + \lambda \bar{c}_t$. $\lambda \bar{c}_t$ is l.s.c. because $\bar{c}_t$ (4) is continuous under A1. $\lambda \bar{c}_t$ is b.b. because $\lambda \bar{c}_t(\chi) \geq -\lambda b > -\infty$ for every $\chi \in \mathbb{X}$. For the second property, note that H_t (9) is continuous for every $t \in \mathbb{T}_N$ because it is either a sum of continuous functions or it is constant. Since F_t is a composition of $\bar{F}_t$ and H_t , it suffices to show that $\bar{F}_t = (f_t^\top, \phi_t^\top, g_t^\top)^\top$ is continuous in each component. The continuity of f_t (1) and ϕ_t (3b) follow from A1. There are two cases for g_t (6). If $t \in \mathbb{T} \setminus \mathbb{T}_y$, then $g_t(y, x; \hat{\theta}) = \hat{\theta}$, which is continuous. Otherwise, $g_t = \bar{g}_t$ (7) depends on a continuous function φ_ϵ , a matrix inverse, and a gradient. The map $(x, \hat{\theta}) \mapsto L_t(x; \hat{\theta})^{-1}$ is continuous because L_t is defined by (8), $L_t(x; \hat{\theta})$ is positive definite for any $(x, \hat{\theta}) \in \mathbb{S} \times \mathbb{P}$, and $\frac{\partial h_t}{\partial \theta}$ is continuous under A1. The gradient $D_\theta \varepsilon_t(y, x; \hat{\theta}) \in \mathbb{R}^p$ is given by

$$D_\theta \varepsilon_t(y, x; \hat{\theta}) = -2 \left(\frac{\partial h_t(x; \hat{\theta})}{\partial \theta} \right)^\top (y - h_t(x; \hat{\theta})). \quad (14)$$

The continuity of $\frac{\partial h_t}{\partial \theta}$ and h_t under A1 imply the continuity of $(y, x, \hat{\theta}) \mapsto D_\theta \varepsilon_t(y, x; \hat{\theta})$. Lastly, the continuity of φ_ϵ and $(y, x, \hat{\theta}) \mapsto L_t(x; \hat{\theta})^{-1} \cdot D_\theta \varepsilon_t(y, x; \hat{\theta})$ imply the continuity of $\bar{g}_t$ (7). $\square$

4. Representation as a regular POMDP

Here, we show that the system (12) equipped with the information-theoretic costs (5) is a nonstationary POMDP with regularity properties under A1. This representation will facilitate policy synthesis in Section 5. Next, we formalize the POMDP model of interest.

Definition 1 (Multi-time-scale Regular POMDP). A multi-time-scale regular POMDP consists of the following: (i) a state space $\mathbb{X}$, compact control space $\mathbb{C}$, and measurement space $\mathbb{Y}$, which are nonempty Borel spaces; (ii) finite discrete-time horizons for the states and measurements, $\mathbb{T}_N$ and $\mathbb{T}_y$, respectively, satisfying the definitions of Section 2; (iii) stage and terminal cost functions c_t that are l.s.c. and b.b. for every $t \in \mathbb{T}_N$; (iv) a distribution $\mathbf{p} \in \mathcal{P}_{\mathbb{X}}$ of the initial state; (v) state transition kernels q_t on $\mathbb{X}$ given $\mathbb{X} \times \mathbb{C}$ that are weakly continuous for every $t \in \{1, 2, \dots, N\}$; and (vi) observation kernels s_t on $\mathbb{Y}$ given $\mathbb{X}$ that are continuous in total variation for every $t \in \mathbb{T}_y$.

Under A1, the system (12) equipped with c_t (5) satisfies Parts (i)–(iv) of Definition 1. We will define state transition and observation kernels and then show that they satisfy Parts (v)–(vi) of

Definition 1 under A1. $\chi_t \in \mathbb{X}$, $u_t \in \mathbb{C}$, and $y_t \in \mathbb{Y}$ are realizations of random objects $\mathcal{X}_t$, U_t , and Y_t , respectively. For every $t \in \mathbb{T}$, the state transition kernel q_{t+1} provides the conditional distribution of $\mathcal{X}_{t+1}$ given a realization of $(\mathcal{X}_t, U_t)$.

Definition 2 (State Transition Kernels $q_1, \dots, q_N$). For every $t \in \mathbb{T}_y \setminus \{N\}$, q_{t+1} depends on the joint distribution ν_t of the process and measurement noise (D_t, W_t) as follows: $q_{t+1}(\mathbb{X} | \chi, u) :=$

$$\nu_t(\{(d, w) \in \mathbb{D} \times \mathbb{Y} : F_t(\chi, u, d, w) \in \mathbb{X}\}), \quad t \in \mathbb{T}_y \setminus \{N\},$$

for every $\mathbb{X} \in \mathcal{B}_{\mathbb{X}}$ and $(\chi, u) \in \mathbb{X} \times \mathbb{C}$. Otherwise, if $t \in \mathbb{T} \setminus \mathbb{T}_y$, then there is no measurement and thus no measurement noise. In this case, q_{t+1} depends on the distribution μ_t of D_t and the function G_t (13):

$$q_{t+1}(\mathbb{X} | \chi, u) := \mu_t(\{d \in \mathbb{D} : G_t(\chi, u, d) \in \mathbb{X}\}), \quad t \in \mathbb{T} \setminus \mathbb{T}_y,$$

for every $\mathbb{X} \in \mathcal{B}_{\mathbb{X}}$ and $(\chi, u) \in \mathbb{X} \times \mathbb{C}$.

q_t differs from a typical state transition kernel in two ways. Its definition accounts for the absence of the measurement noise at particular time points and how this absence leads to dynamics functions with different dependencies. The observation kernel s_t is simpler because it only pertains to $t \in \mathbb{T}_y$. For every $t \in \mathbb{T}_y$, s_t provides the conditional distribution of Y_t given a realization of $\mathcal{X}_t$.

Definition 3 (Observation Kernel s_t for $t \in \mathbb{T}_y$). s_t depends on the distribution ρ_t of W_t as follows:

$$s_t(\mathbb{Y} | \chi) := \rho_t(\{w \in \mathbb{Y} : h_t(x; \hat{\theta}) + w \in \mathbb{Y}\}), \quad t \in \mathbb{T}_y,$$

for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$ and $\chi = (x^\top, \xi^\top, \hat{\theta}^\top)^\top \in \mathbb{X}$.

In the POMDP literature, the observation kernel commonly depends on u_t . We do not allow s_t to depend on u_t to simplify the output sensitivity matrix (2b).

Lemma 1 (Regularity of q_t and s_t). *Let A1 hold. Then, q_t of Definition 2 and s_t of Definition 3 satisfy Part (v) and Part (vi) of Definition 1, respectively.*

Proof. For every $t \in \mathbb{T}$, the weak continuity of q_{t+1} follows from the continuity of F_t (10) under A1 (Theorem 1). For every continuous bounded $\psi : \mathbb{X} \rightarrow \mathbb{R}$, $(\chi, u) \in \mathbb{X} \times \mathbb{C}$, and $t \in \mathbb{T}$, we have that $\int_{\mathbb{X}} \psi \, dq_{t+1}(\cdot | \chi, u) =$

$$\begin{cases} \int_{\mathbb{D} \times \mathbb{Y}} \psi(F_t(\chi, u, \cdot, \cdot)) \, d\nu_t, & \text{if } t \in \mathbb{T}_y \setminus \{N\}, \\ \int_{\mathbb{D}} \psi(G_t(\chi, u, \cdot)) \, d\mu_t, & \text{if } t \in \mathbb{T} \setminus \mathbb{T}_y, \end{cases} \quad (15)$$

by a change-of-measure theorem (Ash, 1972, Th. 1.6.12). Now, let $(\chi_n, u_n)_{n \in \mathbb{N}}$ in $\mathbb{X} \times \mathbb{C}$ converge to $(\chi, u) \in \mathbb{X} \times \mathbb{C}$, and let $\psi : \mathbb{X} \rightarrow \mathbb{R}$ be continuous and bounded. It suffices to show that $\int_{\mathbb{X}} \psi \, dq_{t+1}(\cdot | \chi_n, u_n) \rightarrow \int_{\mathbb{X}} \psi \, dq_{t+1}(\cdot | \chi, u)$ in $\mathbb{R}$ as $n \rightarrow +\infty$ (Rmk. 1). First, consider $t \in \mathbb{T}_y \setminus \{N\}$. Define the function $\psi_{t,n} : \mathbb{D} \times \mathbb{Y} \rightarrow \mathbb{R}$ by $\psi_{t,n} := \psi(F_t(\chi_n, u_n, \cdot, \cdot))$ for every $n \in \mathbb{N}$, and define $\psi_t : \mathbb{D} \times \mathbb{Y} \rightarrow \mathbb{R}$ by $\psi_t := \psi(F_t(\chi, u, \cdot, \cdot))$. We have that $\int_{\mathbb{D} \times \mathbb{Y}} \psi_{t,n} \, d\nu_t = \int_{\mathbb{X}} \psi \, dq_{t+1}(\cdot | \chi_n, u_n)$ for every $n \in \mathbb{N}$ and $\int_{\mathbb{D} \times \mathbb{Y}} \psi_t \, d\nu_t = \int_{\mathbb{X}} \psi \, dq_{t+1}(\cdot | \chi, u)$ by the first case of (15). Since $(\chi_n, u_n) \rightarrow (\chi, u)$ and $\psi \circ F_t$ is continuous, $\psi_{t,n} \rightarrow \psi_t$ pointwise as $n \rightarrow +\infty$. Also, ψ is bounded, and ψ_t and $\psi_{t,n}$ for every $n \in \mathbb{N}$ are continuous and hence measurable. Thus, we use the Dominated Convergence Theorem to conclude that $\int_{\mathbb{D} \times \mathbb{Y}} \psi_{t,n} \, d\nu_t \rightarrow \int_{\mathbb{D} \times \mathbb{Y}} \psi_t \, d\nu_t$ in $\mathbb{R}$ as $n \rightarrow +\infty$ (Ash, 1972, p. 49). The analysis of q_{t+1} for any $t \in \mathbb{T} \setminus \mathbb{T}_y$ is similar, and so, we omit it. For every $t \in \mathbb{T}_y$, the continuity of s_t in total variation is due to h_t being continuous, the measurement noise being additive, ρ_t admitting a continuous nonnegative density, and Scheffé's Lemma (Williams, 1991, Sec. 5.10). The reader may see Kara et al. (2019, Sec. 2.2 (ii)) or (Chapman et al., 2022, Remark 5) for more details. $\square$

Hence, the system (12) with the costs (5), state transition kernels of Definition 2, and observation kernels of Definition 3 is a POMDP that satisfies Definition 1 under A1.

5. Policy synthesis

A POMDP can be reduced to a fully observable *belief-space* MDP, whose state space is the space of posterior distributions of the unobservable state (Rhenius, 1974; Yushkevich, 1976). Moreover, if there exists an optimal policy for the belief-space MDP, then an optimal policy for the POMDP can be constructed (Bertsekas & Shreve, 1996, Prop. 10.3) (Hernández-Lerma, 1989, pp. 89–90). Thus, the purpose of this section is to show the existence of an optimal policy under A1 for a belief-space MDP corresponding to the POMDP of the previous section. (Verifying the equivalence between the POMDP and belief-space MDP solutions is out of scope of the current brief paper; related proofs can be found in Bertsekas and Shreve (1996, Prop. 10.4, Prop. 10.5), for example.)

First, we formalize our belief-space MDP. We define its state and trajectory spaces (Definition 4), random states and controls (Definition 5), random cost variable (Definition 6), Markov policy class (Definition 7), state transition kernels (Definition 8, Definition 9), and initial distribution (Definition 10). Then, we provide a policy existence result (Theorem 2). The definitions are required to state the theorem. While the definitions resemble the standard ones for POMDPs (Bertsekas & Shreve, 1996, Ch. 10) (Feinberg et al., 2016) (Hernández-Lerma, 1989, Ch. 4), we must also circumvent the issue of measurements not necessarily being available always. This issue requires us to provide different definitions for the state transition kernels (Definition 9) and for the initial distribution (Definition 10).

Definition 4 ($\mathbb{Z}, \Omega$). The state space of the belief-space MDP is $\mathbb{Z} := \mathcal{P}_{\mathbb{X}}$. A trajectory $\omega \in \Omega := (\mathbb{Z} \times \mathbb{C})^N \times \mathbb{Z}$ takes the form $\omega = (z_0, u_0, \dots, z_{N-1}, u_{N-1}, z_N)$, where the coordinates of ω are related casually.

Definition 5 ($Z_t, \tilde{U}_t$). We define $Z_t : \Omega \rightarrow \mathbb{Z}$ for every $t \in \mathbb{T}_N$ and $\tilde{U}_t : \Omega \rightarrow \mathbb{C}$ for every $t \in \mathbb{T}$ to be projections, i.e., $Z_t(\omega) := z_t$ and $\tilde{U}_t(\omega) := u_t$ for every $\omega \in \Omega$ of the form in Definition 4.

Definition 6 (C). Under A1, we define $C : \Omega \rightarrow \mathbb{R}^*$ by $C(\omega) := \tilde{c}_N(z_N(\omega)) + \sum_{t=0}^{N-1} \tilde{c}_t(Z_t(\omega), \tilde{U}_t(\omega))$, where $\tilde{c}_t$ is related to c_t (5) as follows:

$$\tilde{c}_t(z_t, u_t) := \int_{\mathbb{X}} c_t(\chi_t, u_t) z_t(d\chi_t), \quad t \in \mathbb{T}, \quad (16a)$$

$$\tilde{c}_N(z_N) := \int_{\mathbb{X}} c_N(\chi_N) z_N(d\chi_N), \quad (16b)$$

with $\tilde{c}_t : \mathbb{Z} \times \mathbb{C} \rightarrow \mathbb{R}^*$ for every $t \in \mathbb{T}$ and $\tilde{c}_N : \mathbb{Z} \rightarrow \mathbb{R}^*$.

Definition 7 ($\Pi, \mathbb{C}_t$). We define $\mathbb{C}_t := \mathbb{C}$ if $t \in \mathbb{T}_u$ and $\mathbb{C}_t := \{\mathbf{u}\}$ if $t \in \mathbb{T} \setminus \mathbb{T}_u$. A control policy $\pi = (\pi_0, \pi_1, \dots, \pi_{N-1}) \in \Pi$ is a finite sequence of measurable stochastic kernels on $\mathbb{C}$ given $\mathbb{Z}$ such that $\pi_t(\mathbb{C}_t | z_t) = 1$ for every $z_t \in \mathbb{Z}$ and $t \in \mathbb{T}$. Π is the set of all policies.

In Definitions 8 and 9, we specify the state transition kernels $\tilde{q}_1, \tilde{q}_2, \dots, \tilde{q}_N$ of the belief-space MDP. For every $t \in \mathbb{T}$, $\tilde{q}_{t+1}(\cdot | z_t, u_t) \in \mathcal{P}_{\mathbb{Z}}$ is the conditional distribution of Z_{t+1} given a realization $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$ of $(Z_t, \tilde{U}_t)$. Definition 8 applies when a measurement is present, whereas Definition 9 applies when a measurement is absent. In each case, some technical lead-up is required before writing the expression for $\tilde{q}_{t+1}$.

Definition 8 ($\tilde{q}_{t+1}$ when a measurement is present). If $t + 1 \in \mathbb{T}_y \setminus \{0\}$, then $\tilde{q}_{t+1}$ is constructed using a joint conditional distribution $r_{t+1}(\cdot, \cdot | z_t, u_t)$ of $(Y_{t+1}, \mathcal{X}_{t+1})$. r_{t+1} is a measurable stochastic kernel on $\mathbb{Y} \times \mathbb{X}$ given $\mathbb{Z} \times \mathbb{C}$ such that $r_{t+1}(\mathbb{Y} \times \mathbb{X} | z_t, u_t) :=$

$$\int_{\mathbb{X}} \int_{\mathbb{Y}} s_{t+1}(\mathbb{Y} | \chi_{t+1}) q_{t+1}(d\chi_{t+1} | \chi_t, u_t) z_t(d\chi_t) \quad (17)$$

for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$, $\mathbb{X} \in \mathcal{B}_{\mathbb{X}}$, and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$. Denote the marginal of r_{t+1} on $\mathbb{Y}$ by r'_{t+1} , i.e., $r'_{t+1}(\mathbb{Y} | z_t, u_t) := r_{t+1}(\mathbb{Y} \times \mathbb{X} | z_t, u_t)$ for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$ and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$. r_{t+1} enjoys a decomposition in terms of this marginal as follows: for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$, $\mathbb{X} \in \mathcal{B}_{\mathbb{X}}$, and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$,

$$r_{t+1}(\mathbb{Y} \times \mathbb{X} | z_t, u_t) = \int_{\mathbb{Y}} \Phi_{t+1}(\mathbb{X} | z_t, u_t, y) r'_{t+1}(dy | z_t, u_t),$$

where Φ_{t+1} is a measurable stochastic kernel on $\mathbb{X}$ given $\mathbb{Z} \times \mathbb{C} \times \mathbb{Y}$ (Bertsekas & Shreve, 1996, Cor. 7.27.1). Then, we define $\tilde{q}_{t+1}(\mathbb{Z} | z_t, u_t) :=$

$$r'_{t+1}(\{y_{t+1} \in \mathbb{Y} : \Phi_{t+1}(\cdot | z_t, u_t, y_{t+1}) \in \mathbb{Z}\} | z_t, u_t) \quad (18)$$

for every $\mathbb{Z} \in \mathcal{B}_{\mathbb{Z}}$ and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$.

Definition 9 ($\tilde{q}_{t+1}$ when a measurement is absent). If $t + 1 \in \{1, 2, \dots, N\} \setminus \mathbb{T}_y$, then $\tilde{q}_{t+1}$ concentrates the realizations of Z_{t+1} at a prior conditional distribution $\eta_{t+1}(\cdot | z_t, u_t) \in \mathcal{P}_{\mathbb{Z}}$ of $\mathcal{X}_{t+1}$; i.e., we define $\tilde{q}_{t+1}$ by

$$\tilde{q}_{t+1}(\mathbb{Z} | z_t, u_t) := \delta_{\eta_{t+1}(\cdot | z_t, u_t)}(\mathbb{Z}) \quad (19)$$

for every $\mathbb{Z} \in \mathcal{B}_{\mathbb{Z}}$ and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$, where η_{t+1} is a measurable stochastic kernel on $\mathbb{Z}$ given $\mathbb{Z} \times \mathbb{C}$, which is defined by Bertsekas and Shreve (1996, p. 261, Eq. (50))

$$\eta_{t+1}(\mathbb{Z} | z_t, u_t) := \int_{\mathbb{X}} q_{t+1}(\mathbb{Z} | \chi_t, u_t) z_t(d\chi_t) \quad (20)$$

for every $\mathbb{Z} \in \mathcal{B}_{\mathbb{Z}}$ and $(z_t, u_t) \in \mathbb{Z} \times \mathbb{C}$.

For $t + 1 \in \{1, 2, \dots, N\}$, the expression for $\tilde{q}_{t+1}$ is in (18) or (19). Lastly, we specify the initial kernel $\tilde{q}_0$, whose construction also depends on the availability of a measurement. $\tilde{q}_0(\cdot | \mathbf{p}) \in \mathcal{P}_{\mathbb{Z}}$ is the distribution of Z_0 when $\mathbf{p} \in \mathbb{Z}$ is the distribution of $\mathcal{X}_0$.

Definition 10 ($\tilde{q}_0$). If $0 \in \mathbb{T}_y$, then $\tilde{q}_0$ depends on a joint distribution $r_0(\cdot, \cdot | \mathbf{p})$ of $(Y_0, \mathcal{X}_0)$, where r_0 is defined by

$$r_0(\mathbb{Y} \times \mathbb{X} | \mathbf{p}) := \int_{\mathbb{X}} s_0(\mathbb{Y} | \chi_0) \mathbf{p}(d\chi_0) \quad (21)$$

$$= \int_{\mathbb{Y}} \Phi_0(\mathbb{X} | \mathbf{p}, y_0) r'_0(dy_0 | \mathbf{p}) \quad (22)$$

for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$, $\mathbb{X} \in \mathcal{B}_{\mathbb{X}}$, and $\mathbf{p} \in \mathbb{Z}$, r'_0 is defined by $r'_0(\mathbb{Y} | \mathbf{p}) := r_0(\mathbb{Y} \times \mathbb{X} | \mathbf{p})$ for every $\mathbb{Y} \in \mathcal{B}_{\mathbb{Y}}$ and $\mathbf{p} \in \mathbb{Z}$, and Φ_0 is a measurable stochastic kernel on $\mathbb{X}$ given $\mathbb{Z} \times \mathbb{Y}$ (Bertsekas & Shreve, 1996, Cor. 7.27.1). However, if $0 \notin \mathbb{T}_y$, then $\tilde{q}_0(\cdot | \mathbf{p})$ concentrates the realizations of Z_0 at $\mathbf{p}$. All together, we define $\tilde{q}_0(\mathbb{Z} | \mathbf{p}) :=$

$$\begin{cases} r'_0(\{y_0 \in \mathbb{Y} : \Phi_0(\cdot | \mathbf{p}, y_0) \in \mathbb{Z}\} | \mathbf{p}), & \text{if } 0 \in \mathbb{T}_y, \\ \delta_{\mathbf{p}}(\mathbb{Z}), & \text{if } 0 \notin \mathbb{T}_y, \end{cases} \quad (23)$$

for every $\mathbb{Z} \in \mathcal{B}_{\mathbb{Z}}$ and $\mathbf{p} \in \mathbb{Z}$.

Equipped with the above definitions, now we can specify a probability measure, an expected cost, and the optimal control problem of interest. By the Ionescu-Tulcea Theorem, for every $\pi \in \Pi$ and $\mathbf{p} \in \mathbb{Z}$, there is a unique probability measure $P_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi} \in \mathcal{P}_{\Omega}$ that depends on π , $\tilde{q}_0(\cdot | \mathbf{p})$, and $\tilde{q}_{t+1}$ for all $t \in \mathbb{T}$. If $G : \Omega \rightarrow \mathbb{R}^*$ is measurable and b.b., then the expectation of G with respect to $P_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi}$ is defined by $E_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi}(G) := \int_{\Omega} G dP_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi}$. The next result guarantees that an optimal policy $\pi^* \in \Pi$ for the belief-space MDP exists under A1, i.e.,

$$E_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi^*}(C) = J^*(\mathbf{p}) := \inf_{\pi \in \Pi} E_{\tilde{q}_0(\cdot | \mathbf{p})}^{\pi}(C), \quad \mathbf{p} \in \mathbb{Z}. \quad (24)$$

Theorem 2 (Policy Synthesis). *Let A1 hold. Define $J_N := \tilde{c}_N$ (16b), and for $t = N - 1, \dots, 1, 0$, define $J_t : \mathbb{Z} \rightarrow \mathbb{R}^*$ recursively backwards in time by*

$$J_t(z_t) := \inf_{u_t \in \mathbb{C}_t} V_t(z_t, u_t), \quad (25)$$

where $\mathbb{C}_t$ is from Definition 7, $V_t := \tilde{c}_t + V'_t$, $\tilde{c}_t$ is defined by (16a), and $V'_t : \mathbb{Z} \times \mathbb{C} \rightarrow \mathbb{R}^*$ depends on J_{t+1} as follows:

$$V'_t(z_t, u_t) := \int_{\mathbb{Z}} J_{t+1}(z_{t+1}) \tilde{q}_{t+1}(dz_{t+1}|z_t, u_t). \quad (26)$$

Then, for every $t \in \mathbb{T}_N$, J_t is l.s.c. and b.b. Also, for every $t \in \mathbb{T}$, there is a measurable function $\kappa_t^* : \mathbb{Z} \rightarrow \mathbb{C}$ such that $\kappa_t^*(z_t) \in \mathbb{C}_t$ and $J_t(z_t) = V_t(z_t, \kappa_t^*(z_t))$ for every $z_t \in \mathbb{Z}$. In particular, for every $t \in \mathbb{T} \setminus \mathbb{T}_u$, we choose $\kappa_t^*(z_t) := \mathbf{u}$ for every $z_t \in \mathbb{Z}$. Lastly, $\pi^* := (\delta_{\kappa_0^*}, \delta_{\kappa_1^*}, \dots, \delta_{\kappa_{N-1}^*}) \in \Pi$ (Rmk. 4) is optimal for the belief-space MDP, i.e., π^* satisfies (24).

Proof. We will prove the first two statements by induction. The arguments have nuances due to the two cases for $\tilde{q}_{t+1}$ (18) (19). The last statement holds by a typical dynamic programming argument, e.g., see Bertsekas and Shreve (1996, Ch. 8) or (Hernández-Lerma & Lasserre, 1996, Ch. 3).

Under A1, for every $t \in \mathbb{T}_N$, $\tilde{c}_t$ (16) is l.s.c. and b.b. because c_t (5) enjoys these properties (Theorem 1) and due to the Generalized Fatou's Lemma (Feinberg et al., 2016, Lemma 6.1). Now, $J_N = \tilde{c}_N$ is l.s.c. and b.b. Assume the induction hypothesis: for some $t \in \mathbb{T}$, J_{t+1} is l.s.c. and b.b. If V'_t (26) is l.s.c. and b.b., then J_t (25) is l.s.c. and b.b., and there is a measurable function κ_t^* that satisfies the properties specified in the statement of Theorem 2 (A1, Rmk. 5). If $\tilde{q}_{t+1}$ is weakly continuous, then V'_t (26) is l.s.c. and b.b. (induction hypothesis, Rmk. 3).

Hence, it suffices to show that $\tilde{q}_{t+1}$ is weakly continuous. First, consider $t + 1 \in \mathbb{T}_y \setminus \{0\}$. Then, q_{t+1} (Definition 2) is weakly continuous, and s_{t+1} (Definition 3) is continuous in total variation under A1 (Lemma 1). These two continuity properties imply that $\tilde{q}_{t+1}$ (18) is weakly continuous directly from Kara et al. (2019, Th. 1), which first appeared in Feinberg et al. (2016). Otherwise, if $t + 1 \in \{1, 2, \dots, N\} \setminus \mathbb{T}_y$, then $\tilde{q}_{t+1}$ is defined by (19). We must show that $(z, u) \mapsto \tilde{q}_{t+1}(\cdot|z, u) = \delta_{\eta_{t+1}(\cdot|z, u)}$ is weakly continuous. We define $\ell : \mathbb{Z} \rightarrow \mathcal{P}_{\mathbb{Z}}$ by $\ell(z) := \delta_z$ and $v : \mathbb{Z} \times \mathbb{C} \rightarrow \mathbb{Z}$ by $v(z, u) := \eta_{t+1}(\cdot|z, u)$, and thus, $\tilde{q}_{t+1}(\cdot|z, u) = \ell(v(z, u))$. Also, ℓ is weakly continuous (Rmk. 3). Hence, it suffices to show that v is weakly continuous. More specifically, it suffices to show that for any continuous bounded function $\psi : \mathbb{X} \rightarrow \mathbb{R}$, the function $\psi' : \mathbb{Z} \times \mathbb{C} \rightarrow \mathbb{R}$ defined by $\psi'(z, u) := \int_{\mathbb{X}} \psi \, d\eta_{t+1}(\cdot|z, u)$ is continuous (Rmk. 1). By the definition of η_{t+1} (20), we have that

$$\psi'(z, u) = \int_{\mathbb{X}} \int_{\mathbb{X}} \psi(a) q_{t+1}(da|\chi, u) z(d\chi). \quad (27)$$

We will re-express ψ' (27) to apply the weak continuity of products of measures in $\mathcal{P}_{\mathbb{X}}$ and $\mathcal{P}_{\mathbb{C}}$. Define $\Psi : \mathbb{X} \times \mathbb{C} \times \mathbb{X} \rightarrow \mathbb{R}$ by $\Psi(\chi, u, \cdot) := \psi$ for all $(\chi, u) \in \mathbb{X} \times \mathbb{C}$. Next, define $\vartheta : \mathbb{X} \times \mathbb{C} \rightarrow \mathbb{R}$ by

$$\vartheta(\chi, u) := \int_{\mathbb{X}} \Psi(\chi, u, a) q_{t+1}(da|\chi, u), \quad (28)$$

which is continuous and bounded (Lemma 1, Rmk. 3). Now, ψ' (27) can be expressed in terms of ϑ (28):

$$\psi'(z, u) = \int_{\mathbb{X}} \int_{\mathbb{C}} \vartheta(\chi, \bar{u}) \delta_u(d\bar{u}) z(d\chi) \quad (29)$$

$$= \int_{\mathbb{X} \times \mathbb{C}} \vartheta \, d(z\delta_u), \quad (30)$$

where $z\delta_u \in \mathcal{P}_{\mathbb{X} \times \mathbb{C}}$ is the product of $z \in \mathcal{P}_{\mathbb{X}}$ and $\delta_u \in \mathcal{P}_{\mathbb{C}}$.

Let $z_n \rightarrow z$ weakly in $\mathbb{Z}$, and let $u_n \rightarrow u$ in $\mathbb{C}$. The latter implies that $\delta_{u_n} \rightarrow \delta_u$ weakly in $\mathcal{P}_{\mathbb{C}}$ (Rmk. 3). The weak convergence of $z_n \rightarrow z$ in $\mathcal{P}_{\mathbb{X}}$ and $\delta_{u_n} \rightarrow \delta_u$ in $\mathcal{P}_{\mathbb{C}}$ implies the weak convergence of $z_n \delta_{u_n} \rightarrow z\delta_u$ in $\mathcal{P}_{\mathbb{X} \times \mathbb{C}}$ (Rmk. 3). The weak convergence of these products is equivalent to the convergence of $\int_{\mathbb{X} \times \mathbb{C}} h \, d(z_n \delta_{u_n}) \rightarrow$

$\int_{\mathbb{X} \times \mathbb{C}} h \, d(z\delta_u)$ in $\mathbb{R}$ for every continuous bounded function $h : \mathbb{X} \times \mathbb{C} \rightarrow \mathbb{R}$ (Rmk. 1). By considering $h = \vartheta$ (28), we find that $\psi'(z_n, u_n) \rightarrow \psi'(z, u)$ in $\mathbb{R}$ in view of (30). $\square$

This concludes the theoretical portion of the brief paper.

6. Conceptual example

In this section, we provide a conceptual leukemia treatment model and explain why it satisfies A1. We consider the setting of adjusting the dose of an oral chemotherapy called 6-Mercaptopurine (6-MP) during a treatment cycle for a leukemia patient. Currently, the dose is adjusted according to the current neutrophil concentration (a neutrophil is a type of white blood cell). The concentration is measured at least once per cycle, whose typical length is 21 days. If the concentration is outside a desired range, then the dose of 6-MP is modified (increased or decreased by 20%) for the next cycle. Nominally, the drug is taken at the prescribed dose for the first 14 days, and then no drug is taken for the remaining 7 days.

Continuous-time models for the kinetics of 6-MP (Jayachandran et al., 2015; Jayachandran, Rundell, Hannemann, Vik, & Ramkrishna, 2014) and their effect on white blood cells (Jayachandran et al., 2014; Karppinen, Lohi, & Vihola, 2019) have been developed recently. The models are deterministic in Jayachandran et al. (2015) (Jayachandran et al., 2014). Three states represent the amounts of 6-MP-derived substances in Jayachandran et al. (2015). Five states represent the amounts of white blood cells in different stages of maturity under the influence of 6-MP in Jayachandran et al. (2014). A stochastic model with two states that neglects the delayed effect of 6-MP on white blood cells has been proposed in Karppinen et al. (2019). These models represent one or both of the following biochemical processes: (1) how 6-MP is broken down into a substance called 6-thioguanine nucleotide (6-TGN); (2) how 6-TGN influences the proliferation and maturation of white blood cells in the bone marrow.

Building on the above models, we consider a discrete-time stochastic model for the two biochemical processes. The length of $\mathbb{T}$ is the length of a typical cycle (21 days). $\mathbb{T}_u$ equals the time points corresponding to the first 14 days. If weekly measurements are planned, then $\mathbb{T}_y$ equals the time points corresponding to days 0, 7, and 14. Fig. 2 illustrates the different time scales. The control space $\mathbb{C} = [0, \bar{u}] \subset \mathbb{R}$ is compact (A1 (c)), where $\bar{u} \in \mathbb{R}_+$ is the maximum permissible dose rate. We may choose $\bar{u}$ (mg/day) to be twice the nominal dose rate of 50 (mg/(m² · day)), i.e., $\bar{u} = 2 \cdot 50 \cdot \text{BSA}$, where BSA (m²) is the patient's body surface area. The default value is $\mathbf{u} = 0$, which refers to no drug being taken. We assume Gaussian process and measurement noise due to the general role of noise in biochemical systems (Eling et al., 2019; Rao et al., 2002) and lack of domain-specific knowledge about the noise distributions. This is one way to satisfy the measurement noise condition (A1 (e)). In our model, x_1, x_2 , and x_3 are the three states representing the amounts of 6-MP-derived substances from Jayachandran et al. (2015), while x_4 through x_8 are the five states representing the amounts of white blood cells in different stages of maturity under the influence of 6-MP from Jayachandran et al. (2014) (Table 1). The parameters include various rates, drug-effect quantities, and the ratio between neutrophils and mature white blood cells (Table 1). The measurements are neutrophils and mature white blood cells. The spaces of interest are $\mathbb{S} = \mathbb{R}_+^8$, $\mathbb{P} = \mathbb{R}_+^{13}$, $\mathbb{C} = [0, \bar{u}]$, $\mathbb{D} = \mathbb{R}^8$, and $\mathbb{Y} = \mathbb{R}^2$.

Letting $\Delta \in \mathbb{R}_+$ be the duration of $[t, t + 1)$ for every $t \in \mathbb{T}$ and $\hat{\varphi}_\epsilon : \mathbb{R}^8 \rightarrow \mathbb{S}$ be a continuously differentiable approximation

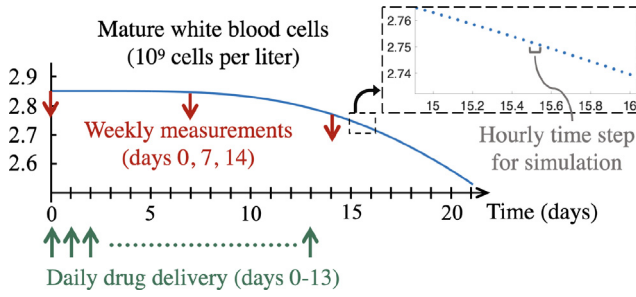


Fig. 2. An illustration of the time scales in the conceptual leukemia treatment example. The close-up illustrates $\mathbb{T}$, where the time steps are hourly. For the illustration, we have simulated the mature white blood cells using a nominal model (to be presented in Eq. (33)) and the dosing regimen from DeAngelo et al. (2015). In the simulation, we have initialized the states to be the equilibrium values of the dynamics when the drug input is zero; parameter values are from Jayachandran et al. (2015, 2014).

Table 1
Description of states and parameters for the leukemia treatment model.

Symbol	Description
x_1	Amount of 6-MP in the gut
x_2	Amount of 6-MP in the plasma
x_3	Concentration of 6-TGN in red blood cells
x_4	Concentration of proliferating white blood cells
x_5, x_6, x_7	Concentrations of maturing white blood cells
x_8	Concentration of mature white blood cells
θ_1	Conversion rate (6-MP to 6-TGN)
θ_2	Michaelis–Menten constant
θ_3	Maximum proliferation rate
θ_4	Steepness parameter
θ_5	Feedback parameter
θ_6	Maximum drug effect on mature white blood cells
θ_7	Saturation constant for drug effect
$\theta_8, \dots, \theta_{12}$	Rate of absorption, elimination, or transition
θ_{13}	Ratio between neutrophils and mature white blood cells

for $\varphi_\epsilon : \mathbb{R}^8 \rightarrow \mathbb{S}$ (Section 1.2),³ we consider

$$f_t(x, u, d; \theta) = \hat{\varphi}_\epsilon(x + \Delta \cdot \bar{f}(x, u; \theta) + d), \quad (31)$$

$$h_t(x; \theta) = C(\theta)x, \quad (32)$$

$$\bar{f}(x, u; \theta) := A(\theta)x + Bu + \hat{f}(x; \theta), \quad (33)$$

where $A(\theta) \in \mathbb{R}^{8 \times 8}$ and $C(\theta) \in \mathbb{R}^{2 \times 8}$ depend linearly on θ , $B \in \mathbb{R}^8$ is constant, and $\hat{f}$ is nonlinear. The definition of $\bar{f}$ (33) comes from Jayachandran et al. (2015) and Jayachandran et al. (2014). The nonlinear part is $\hat{f} := [0, \hat{f}_2, -\hat{f}_2, \hat{f}_4, 0, 0, 0, 0]^\top$, where

$$\hat{f}_2(x; \theta) := -\theta_1 x_2 (\theta_2 + x_2)^{-1}, \quad (34)$$

$$\hat{f}_4(x; \theta) := \left(\frac{\theta_3}{1 + (x_8/\theta_5)^{\theta_4}} - \frac{\theta_6 x_3}{\theta_7 + x_3} \right) x_4. \quad (35)$$

³ One choice is $\max\{x_i, \epsilon\} \approx \hat{\varphi}_{\epsilon, \beta}(x_i) := \beta^{-1} \log(\exp(\beta x_i) + \exp(\beta \epsilon))$ with $\beta \geq 1$, which satisfies $\max\{x_i, \epsilon\} \leq \hat{\varphi}_{\epsilon, \beta}(x_i) \leq \max\{x_i, \epsilon\} + \beta^{-1} \log(2)$.

$\hat{f}_2$ describes how 6-MP in the plasma (x_2) is broken down into 6-TGN (x_3). $\hat{f}_4$ describes how the expansion rate of the proliferating white blood cells (x_4) is influenced by the mature white blood cells (x_8) and the drug-derived substance x_3 . The linear part $A(\theta)x + Bu$ is given by

$$A(\theta)x + Bu = \begin{bmatrix} -\theta_8 x_1 + u \\ \theta_8 x_1 - \theta_9 x_2 \\ -\theta_{10} x_3 \\ -\theta_{11} x_4 \\ \theta_{11}(x_4 - x_5) \\ \theta_{11}(x_5 - x_6) \\ \theta_{11}(x_6 - x_7) \\ \theta_{11} x_7 - \theta_{12} x_8 \end{bmatrix}, \quad (36)$$

which includes the ingestion of 6-MP and the proliferation and maturation of the white blood cells. One can show that f_t (31) satisfies Parts (a) and (f) of A1. The amounts of mature white blood cells (x_8) and neutrophils ($\theta_{13} x_8$) are relevant for clinical decisions. We choose $y \in \mathbb{R}^2$, where y_1 and y_2 are the measured amounts of mature white blood cells and neutrophils, respectively, and thus,

$$C(\theta) = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 0 & \theta_{13} \end{bmatrix} \in \mathbb{R}^{2 \times 8}, \quad (37)$$

where $\theta_{13} \in (0, 1)$ is the ratio between neutrophils and mature white blood cells (Table 1). The measurement equation h_t (32) satisfies Parts (d) and (f) of A1.

Lastly, we specify cost functions $\hat{c}_t$ to penalize (i) deviations of the amount of neutrophils with respect to a desired range $[b, \bar{b}]$ and (ii) low doses of 6-MP. The first criterion serves to protect the patient's immune system, while the second criterion serves to limit the production of cancer cells. For example, we may choose

$$\hat{c}_t(x, u; \theta) := \frac{1}{N+1} (\zeta(x; \theta) - \hat{\lambda} u^2), \quad t \in \mathbb{T}, \quad (38a)$$

$$\hat{c}_N(x; \theta) := \frac{1}{N+1} \zeta(x; \theta), \quad (38b)$$

$$\zeta(x; \theta) := (\theta_{13} x_8 - \underline{b})(\theta_{13} x_8 - \bar{b}), \quad (38c)$$

with $\hat{\lambda} \in \mathbb{R}_+$, where ζ penalizes deviations of the amount of neutrophils with respect to $[b, \bar{b}]$. $\hat{c}_t$ (38) is continuous and hence l.s.c. for every $t \in \mathbb{T}_N$. ζ is b.b. since it is convex quadratic. Since $0 \leq u \leq \bar{u}$, we have that $-\hat{\lambda} u^2 \geq -\hat{\lambda} \bar{u}^2$. Thus, $\hat{c}_0, \hat{c}_1, \dots, \hat{c}_N$ (38) satisfy Part (b) of A1.

7. Conclusion

We have provided a theoretical foundation that unifies several features of biomedical systems, e.g., multiple time scales, partial observability, and uncertain parameters. A future theoretical step is to analyze the performance of the control policy from Theorem 2 under the true dynamics (1), which depend on θ rather than $\hat{\theta}_t$. A shortcoming of Section 6 is that the partial observability and high dimensionality of the model (which is based on models from Jayachandran et al., 2015 and Jayachandran et al., 2014) preclude the practical application of Theorem 2. Indeed, POMDPs are notorious for their numerical complexity, except in special cases such as the linear–quadratic–Gaussian case, which does not apply here. Hence, one aim of our ongoing work is the development of methodology with reduced complexity. We are collecting leukemia patient data, which can be useful for reducing the parameter space (i.e., use the data to identify the subset of parameters that are common to the patient population, as in, e.g., Jost et al. (2020)). Taking inspiration from the framework proposed here, we hope to devise new methods that are resilient to process and measurement noise but also are

computationally tractable, which may be facilitated by patient data and domain-specific modeling structure.

In the long term, we envision a technology that provides optimized doses for the next cycle given the patient's prior doses and measurements. During a clinical visit, the patient and her oncologist would see on a computer screen how the predicted trajectories of mature white blood cells and neutrophils refit to the patient's data set when a new measurement becomes available. The optimized doses would be compared to the typical doses visually, and then the oncologist would decide which doses to prescribe for the next cycle. The technology may benefit from the theoretical foundation that we have developed here to conceptually unify the unobservable state, unknown parameters, multiple time scales, and nonlinear stochastic dynamics.

Acknowledgments

M.P.C. appreciates Chuaning Wei, Zhengmin Yang, Zehua Li, and Huizhen Yu for fruitful discussions, and she appreciates support provided by the Edward S. Rogers Sr. Department of Electrical and Computer Engineering, University of Toronto. The authors wish to thank two anonymous reviewers for their helpful suggestions.

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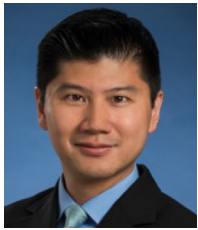
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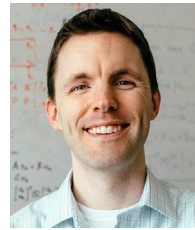


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